

CONFIDENTIAL — PORTFOLIO ARTIFACT

MITRE ATT&CK

THREAT MAPPING

Maplewood Regional Medical Center

Ransomware Attack Chain Analysis

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FRAMEWORK

MITRE ATT&CK v14

DATE

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VERSION

1.0

EXECUTIVE SUMMARY

This threat mapping document models a ransomware attack against **Maplewood Regional Medical Center**, a fictional mid-size healthcare organization. The attack chain is mapped to the **MITRE ATT&CK framework (v14)** across seven tactics and eight techniques, tracing adversary behavior from initial phishing delivery through full network encryption and double extortion. Detection coverage is cross-referenced against a **Wazuh SIEM threat detection lab** using Atomic Red Team simulations, identifying both confirmed detections and critical visibility gaps. This document is paired with an Incident Response Tabletop Exercise (TTX) mapped to **NIST SP 800-61**, covering both the technical attack path and the organizational response.

TACTICS	TECHNIQUES	DETECTIONS	GAPS	COVERAGE
7	8	4	4	50%

SCOPE & METHODOLOGY

Target Organization	Maplewood Regional Medical Center (Fictional Healthcare)
Scenario	Ransomware, Double Extortion
Industry	Healthcare
Primary Framework	MITRE ATT&CK v14
Supporting Framework	NIST SP 800-61 Rev 2 (paired TTX document)
Detection Platform	Wazuh SIEM 4.7.3 using Atomic Red Team simulations on Windows 11 with Sysmon
Companion Document	Incident Response TTX for Maplewood Regional Medical Center

ATTACK CHAIN OVERVIEW

The table below maps the ransomware attack chain against Maplewood Regional Medical Center to MITRE ATT&CK tactics and techniques, with detection status cross-referenced against the Wazuh lab.

#	Tactic	Technique	ATT&CK ID	Maplewood Context	Detection
1	Initial Access	Spearphishing Attachment	T1566.001	Malicious email disguised as patient record triggers payload on staff workstation	✗ Gap
2	Execution	PowerShell	T1059.001	Hidden PowerShell script executes silently after attachment is opened	✓ Detected
3	Persistence	Registry Run Keys	T1547.001	Malware written to registry survives reboots across nursing shift changes	✓ Detected
3	Persistence	Scheduled Task	T1053.005	Timed execution maintains attacker presence independent of user activity	✓ Detected
4	Defense Evasion	Disable or Modify Tools	T1562.001	Windows Defender disabled, exploiting the limited IT staffing at Maplewood	✗ Gap
5	Credential Access	LSASS Memory Dumping	T1003.001	ProcDump harvests staff credentials enabling access to EHR and backup systems	✓ Detected
6	Discovery	Network Service Scanning	T1046	Internal network scanned to map EHR servers, backups, and domain controllers	✗ Gap
7	Impact	Data Encrypted for Impact	T1486	Ransomware deployed across all systems; PHI exfiltrated for double extortion	✗ Gap

TECHNIQUE ANALYSIS

Each technique is analyzed in the context of the Maplewood environment with detection findings and recommended mitigations.

T1566.001: Spearphishing Attachment

INITIAL ACCESS

✗ DETECTION GAP

Wazuh: Not simulated

Context	The attack begins with a targeted phishing email sent to Maplewood hospital staff containing a malicious attachment disguised as a patient record, medical invoice, or clinical document. Hospital employees handle these file types around the clock, making them highly susceptible. The attacker exploits the trust employees place in familiar document formats, relying on limited security awareness training common in healthcare environments.
Detection	T1566.001 was not simulated in the Wazuh lab. Email-based phishing detection requires dedicated email security controls like a secure email gateway and attachment sandboxing, which fall outside SIEM scope. This represents a perimeter visibility gap.
Mitigation	Deploy a Secure Email Gateway with attachment sandboxing. Conduct phishing simulation training for all clinical and administrative staff. Implement DMARC, DKIM, and SPF email authentication controls.

T1059.001: PowerShell

EXECUTION

✓ DETECTED

Wazuh: 4 alerts fired at max Level 15 severity

Context	Following delivery of the phishing attachment, an embedded PowerShell script executes silently on the victim workstation. The employee sees nothing unusual while attacker commands run in the background. PowerShell is a legitimate Windows administration tool, making it a preferred execution vehicle for ransomware operators seeking to blend in with normal system activity.
Detection	DETECTED via Atomic Red Team simulation (T1059.001). Four alerts fired at maximum Level 15 severity. Wazuh rules triggered on suspicious PowerShell execution patterns via Sysmon telemetry.
Mitigation	Enable PowerShell Script Block Logging via Group Policy. Implement AMSI integration. Enforce Constrained Language Mode for non-administrative accounts. Alert on encoded command usage (-EncodedCommand flag).

T1547.001: Registry Run Keys

PERSISTENCE

✓ DETECTED

Wazuh: Rules 92302 and 92307 fired

Context	To survive system reboots across nursing and administrative shift changes, the attacker writes malware to Windows Registry Run Keys. This ensures the malicious payload re-executes automatically each time an employee logs into the compromised workstation, maintaining a persistent foothold without requiring re-establishment of access after each interruption.
Detection	DETECTED via Atomic Red Team simulation (T1547.001). Sysmon auto-tagged the technique and Rules 92302 and 92307 fired. Wazuh correctly identified the registry persistence mechanism.
Mitigation	Monitor and alert on writes to HKLM\Software\Microsoft\Windows\CurrentVersion\Run and related keys. Implement application whitelisting to prevent unauthorized executables from running at startup.

T1053.005: Scheduled Task

PERSISTENCE

✓ DETECTED

Wazuh: Rule 60009 fired

Context	In addition to registry persistence, the attacker creates a scheduled task to execute malicious code at defined intervals. This secondary persistence mechanism operates independently of user login activity, allowing the attacker to maintain execution capability even if the registry-based persistence is detected and remediated.
Detection	DETECTED via Atomic Red Team simulation (T1053.005). Rule 60009 fired on scheduled task creation. Detection confirmed through Windows Event Log telemetry.
Mitigation	Audit scheduled tasks and alert on creation by non-administrative accounts. Monitor schtasks.exe execution via endpoint telemetry. Restrict scheduled task creation permissions across standard user accounts.

T1562.001: Disable or Modify Tools

DEFENSE EVASION

✗ DETECTION GAP

Wazuh: Not simulated

Context	After establishing persistence, the attacker disables Windows Defender and any endpoint security agents on the compromised workstation. This eliminates the primary alerting mechanism and creates an open runway through the network. Maplewood's limited IT staffing makes this technique especially effective. Disabled security tools can go unnoticed for a long time before anyone manually reviews the environment.
Detection	T1562.001 was not simulated in the Wazuh lab. This is a critical detection gap. If endpoint security is disabled, the SIEM loses its primary telemetry source and all subsequent attacker activity becomes invisible.
Mitigation	Implement tamper protection on all endpoint security tools. Configure SIEM alerts for unexpected agent disconnections. Use cloud-managed EDR that cannot be disabled by local administrator accounts.

T1003.001: LSASS Memory Dumping

CREDENTIAL ACCESS

✓ DETECTED

Wazuh: Rule 92032 fired, 93MB LSASS dump confirmed

Context	Using ProcDump, the attacker dumps credentials stored in LSASS memory on the compromised workstation. Valid staff credentials enable authenticated access across Maplewood's internal network including EHR systems, administrative servers, and backup infrastructure. Compromised backup credentials are a key enabler of the double extortion model. If backups are encrypted or deleted, the victim loses their recovery option.
Detection	DETECTED via Atomic Red Team simulation (T1003.001). Rule 92032 fired. A 93MB LSASS memory dump was confirmed via ProcDump. Detection validated through Sysmon process access telemetry.
Mitigation	Enable Windows Credential Guard and Protected Process Light (PPL) for LSASS. Restrict ProcDump and Sysinternals tools via application control policy. Alert on any process attempting to access LSASS memory.

T1046: Network Service Scanning

DISCOVERY

✗ DETECTION GAP

Wazuh: Not simulated

Context	With valid credentials in hand, the attacker conducts internal network reconnaissance to map Maplewood's environment. Active scanning reveals IP addresses, open ports, running services, and the location of high-value targets including EHR servers, backup systems, and domain controllers. This is the attacker's planning stage. They are mapping out exactly which systems to hit for maximum operational and financial damage before they ever pull the trigger on the ransomware.
Detection	T1046 was documented as a detection gap in the Wazuh lab. No compatible Windows 11 Atomic Red Team tests were available. Maplewood would have zero visibility into internal scanning activity on their own network.
Mitigation	Deploy network intrusion detection at internal segment boundaries. Implement network traffic baselining to detect anomalous scan patterns. Use micro-segmentation to limit lateral movement between clinical and administrative networks.

✖ DETECTION GAP

Wazuh: Not simulated

Context	Having mapped the environment and accessed backup systems, the attacker simultaneously deploys ransomware across all identified systems. EHR databases, patient records, billing systems, and backup infrastructure are encrypted in a coordinated detonation. A ransom demand is issued threatening public release of exfiltrated PHI if payment is not received — creating dual pressure under both operational continuity and HIPAA compliance dimensions. Patient care is directly disrupted.
Detection	T1486 was not simulated in the Wazuh lab. Ransomware detonation detection requires behavioral analysis rules tuned for mass file encryption events, which were outside the scope of the current lab configuration.
Mitigation	Implement immutable, air-gapped backups tested regularly for restoration. Deploy behavioral EDR capable of detecting mass file modification. Establish a ransomware-specific IR playbook with pre-defined isolation procedures and HIPAA breach notification workflows.

DETECTION COVERAGE SUMMARY

Cross-referencing the modeled attack chain against Wazuh SIEM detections from the Atomic Red Team lab reveals a **50% detection coverage rate** across the eight mapped techniques. While mid-chain techniques such as Execution, Persistence, and Credential Access are detectable, critical blind spots exist at the entry point, evasion layer, reconnaissance phase, and detonation stage. This coverage profile is representative of many healthcare organizations operating with limited security tooling and staffing.

Technique	ATT&CK ID	Tactic	Status	Wazuh Evidence
Spearphishing Attachment	T1566.001	Initial Access	✗ Gap	—
PowerShell	T1059.001	Execution	✓ Detected	Level 15 — 4 alerts
Registry Run Keys	T1547.001	Persistence	✓ Detected	Rules 92302, 92307
Scheduled Task	T1053.005	Persistence	✓ Detected	Rule 60009
Disable or Modify Tools	T1562.001	Defense Evasion	✗ Gap	—
LSASS Memory Dumping	T1003.001	Credential Access	✓ Detected	Rule 92032
Network Service Scanning	T1046	Discovery	✗ Gap	—
Data Encrypted for Impact	T1486	Impact	✗ Gap	—

KEY FINDINGS & RECOMMENDATIONS

CRITICAL

No visibility at entry point or detonation

Maplewood has no detection capability for phishing attachment delivery (T1566.001) or ransomware detonation (T1486), which are the two highest-impact phases of the chain. Deploy a Secure Email Gateway with sandboxing and behavioral EDR capable of detecting mass file encryption events.

HIGH

Defense evasion eliminates SIEM telemetry

T1562.001 disabling endpoint security removes Wazuh's primary data source. Once those tools go down, all subsequent attacker activity becomes completely invisible. Implement tamper protection and cloud-managed EDR that cannot be disabled by local accounts.

HIGH

No internal network reconnaissance detection

T1046 network scanning is completely undetected, allowing the attacker to map the entire Maplewood environment without triggering any alerts. Deploy network IDS at internal boundaries and implement micro-segmentation between clinical and administrative networks.

MEDIUM	Mid-chain detections confirmed but response is alert-only T1059.001, T1547.001, T1053.005, and T1003.001 are all detected by Wazuh. However, responses are alert-only with no automated containment. Implement automated isolation playbooks so detections trigger response actions without manual analyst intervention.
MEDIUM	PHI exfiltration creates HIPAA breach notification obligation The double extortion model targets PHI exfiltration before encryption. Maplewood faces both ransom pressure and mandatory HIPAA breach notification obligations. Ensure the IR plan includes a legal and compliance notification workflow and pre-engage breach counsel before an incident occurs.

This document is a fictional threat mapping exercise created for cybersecurity portfolio purposes. Maplewood Regional Medical Center is a fictional organization. All techniques reference MITRE ATT&CK v14. Companion document: Incident Response TTX for Maplewood Regional Medical Center, mapped to NIST SP 800-61 Rev 2.